

Audrey Shih

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EDUCATION

Stanford University Ph.D., Chemical Engineering, GPA 3.96 M.S., Chemical Engineering	Stanford, CA Expected June 2025 2020 – 2023
Princeton University B.S.E., Chemical and Biological Engineering, <i>cum laude</i> Minor, Materials Science and Engineering	Princeton, NJ 2016 – 2020

EXPERIENCE

Stanford University , Stanford, CA <i>Doctoral Researcher</i> PI: Gerald G. Fuller	2021 – Present
- Developed and optimized a portable magnetic stress rheometer to enable real-time fluid flow predictions for biopolymeric materials, using CAD-designed components and Raspberry Pi to reduce costs from ~\$90k to \$160. - Engineered 3D bioprinting systems for rheological analysis of bioink gelation in diffusion-driven crosslinking to optimize mechanical properties of printed complex polymeric tissues. - Managed BSL-2 lab safety protocols as designated lab safety officer, attending quarterly environmental health and safety meetings with University officials to ensure regulatory and safety compliance in research environments. - Instructed and mentored both undergrad and grad students in <i>Mechanics of Soft Matter: Rheology (CHEMENG 470)</i>.	
Stanford University , Stanford, CA <i>Doctoral Researcher</i> PI: Joseph M. DeSimone	2021
- Developed and optimized high-resolution 3D Continuous Liquid Interface Production (CLIP) printing through experiments utilizing CAD and 3D-slicing software to test printer settings and SEM imaging. - Simulated optics and photopolymerization transport and kinetics, improving precision and resolution for advanced additive manufacturing processes by building a digital twin. - Co-authored patent on design and manufacturing of 3D-printed lattice microneedle array patches for drug delivery.	
Princeton University , Princeton, NJ <i>Undergraduate Researcher</i> PI: Sujit S. Datta	2018 – 2020
- Investigated elastic turbulence in polymer flow by CAD designing 3D-printed microfluidic model porous channels. - Fluorescently tagged DNA to visualize polymer configurational changes in flow fields using microscopy. - Featured in ACEE's first ever spotlight article on undergraduate contributions to environmental studies. - Won Outstanding Thesis Award, thesis featured in profile article by the School of Engineering and Applied Science.	
TU-Dortmund, Fakultät Bio und Chemieingenieurwesen , Dortmund, Germany <i>Research Intern</i> PI: Armin Quentmeier	2018
- Commandeered the Biochemical and Chemical Engineering department's contribution to the 2018 Engineering Meets Art exhibition, designed project involving photographs of microorganisms and colonies. - Wrote 40-page manual of protocols for inoculation and dilution techniques, instructions for using cameras, lenses, microscopes, and Adobe Photoshop Elements, and analyzed microscopic images using various software.	

SKILLS

Technical: Equipment troubleshooting; fluid mechanics; materials characterization and synthesis; medical devices; rheology; heat transfer; numerical simulations; statistical analysis; DOE; CAD (Fusion 360, SolidWorks); 3D printing (SLA, DLS, FDM, CLIP); laser cutting; soldering; CNC machining; hardware/embedded systems (PCB assembly, Raspberry Pi, Peltier module); SEM; confocal microscopy; fluorescence staining; FTIR; GPC; DMA; PCR; cell culture.
Software: Java, MATLAB, Python, R; ImageJ; PIV; ChemDraw; Adobe Illustrator, Photoshop; LaTeX; Microsoft Office.
Languages: fluent in English and Mandarin, proficient in Spanish.

PUBLICATIONS & PATENTS

- Shih, A.**, Chung, S. J., Brunel, L. G., Christakopoulos, F., Heilshorn, S. C., Fuller, G. G., *In situ* rheological measurements of bioink crosslinking. *In preparation*.
- Shih, A.**, Chung, S. J., Shende, O. B., Herwald, S. E., Vezeridis, A. M., Fuller, G. G., [Viscoelastic measurements of abscess fluids using a magnetic stress rheometer](#). *Physics of Fluids* 36 (11): 111919 (2024), Cover Article.

- Cai, P. C., Braunreuther, M., **Shih, A.**, Spakowitz, A. J., Fuller, G. G., Heilshorn, S. C., [Air-liquid intestinal cell culture allows *in situ* rheological characterization of intestinal mucus](#). *APL Bioengineering* 8 (2): 026112 (2024).
- DeSimone, J. M., Jacobson, G. B., Dulay, M. T., Lee, B. J., Hsiao, K., Rajesh, N., Driskill, M. M., **Shih, A.**, et. al., *Polymeric microstructures and systems and methods for making same*. [Patent number WO2023049267A1 \(2023\)](#).
- Hsiao, K., Lee, B. J., Samuelson, T., Lipkowitz, G., Kronenfeld, J. M., Ilyn, D., **Shih, A.**, et al., [Single-digit-micrometer-resolution continuous liquid interface production](#). *Science Advances* 8, eabq2846 (2022).
- Browne, C. A., **Shih, A.**, Datta, S. S., [Bistability in the Unstable Flow of Polymer Solutions in Porous Media](#). *Journal of Fluid Mechanics* 890:A2 (2020).
- Browne, C. A., **Shih, A.**, Datta, S. S., [Pore-Scale Flow Characterization of Polymer Solutions in Microfluidic Porous Media](#). *Small* 16, 1903944 (2019).

SELECTED AWARDS

2024	Stanford University 3 Minute Thesis Competition Honorable Mention
2024	Excellence in Fluid Mechanics Research and Oral Presentation Award, AIChE Annual Meeting
2024, 2020	Sigma Xi Scientific Research Honor Society Nominee
2023	Chemical Engineering Department Service Leadership Award
2021	Judges' Vote and Audience Choice poster awards, Stanford MIPS Retreat, Woodside, CA
2020	National Science Foundation (NSF) Graduate Research Fellowship
2020	Lore von Jaskowsky Memorial Prize, Princeton School of Engineering and Applied Sciences
2020	Materials Science and Engineering Department Outstanding Senior Thesis Award
2019	PSEG Best Poster Award, ACEE Annual Meeting, Princeton, NJ

CONFERENCE TALKS

- **Shih, A.**, Chung, S. J., Shende, O. B., Brunel, L. G., Christakopoulos, F., Herwald, S. E., Vezeridis, A. M., Fuller, G. G., *Magnetic stress rheometer for biological fluid characterization*. American Society of Chemical Engineers Annual Meeting (AIChE 2024), San Diego, CA, Oct. 2024.
- **Shih, A.**, Chung, S. J., Shende, O. B., Herwald, S. E., Vezeridis, A. M., Fuller, G. G., *Rheological Characterization of Biological Fluids*. NETZSCH Introduction to Rheology and Thermal Analysis Workshop, Palo Alto, CA, Oct. 2024.
- **Shih, A.**, Chung, S. J., Herwald, S. E., Vezeridis, A. M., Fuller, G. G., *Magnetic stress rheometer for biological fluid characterization*. American Society of Chemical Engineers Annual Meeting (AIChE 2023), Orlando, FL, Nov. 2023.
- **Shih, A.**, Chung, S. J., Herwald, S. E., Vezeridis, A. M., Fuller, G. G., *Magnetic stress rheometer for biological fluid characterization*. XIXth International Congress on Rheology (ICR 2023), Athens, Greece, Aug. 2023.
- **Shih, A.**, Chung, S. J., Herwald, S. E., Vezeridis, A. M., Fuller, G. G., *Magnetic stress rheometer for biological fluid characterization*. Society of Rheology Annual Meeting (SOR 2022), Chicago, IL, Oct. 2022.

LEADERSHIP & MENTORSHIP

Stanford University, Stanford, CA 2022 – Present
Research Mentor

- Mentored one visiting Ph.D. student (University of Tokyo), two undergraduate students (Stanford University, Shenzhen University), and one high school student (Germantown Friends School), teaching CAD, Python, MATLAB, ImageJ, fluid mechanics, and polymer theory.
- Provided hands-on guidance in hardware, coding, and data analysis, fostering technical skills and problem-solving abilities for real-world research projects in rheology, medicine, and 3D bioprinting.

Graduate Life Office, Stanford University, Stanford, CA 2021 – Present
Community Associate

- Coordinated 12 events annually to foster community among graduate student population, managing a \$3,600 budget for diverse and inclusive programs, including welcome, orientation, and multicultural initiatives.
- Advised graduate residents on support services and professional development, utilizing annual training on suicide prevention, servant leadership, bias intervention, and sexual harassment prevention.

Graduate Student Action Committee, Stanford University, Stanford, CA 2022 – 2023
Social Events Chair | Department of Chemical Engineering

- Served as liaison between ChemE student community and administration through quarterly meetings.
- Spearheaded multi-departmental events to expand the department's community, promoting social and collaborative connections across various department programs.

Stanford University, Stanford, CA

2022 – 2023

First-Year Mentor | Department of Chemical Engineering

- Guided first-year students in selecting labs, managing coursework, and preparing for qualification exams, sharing resources and support through three meetings per quarter.

Stanford Splash, Stanford, CA

2023

Teacher Volunteer

- Taught hands-on workshop on polymer crosslinking to middle- and high-school students in two classes.

PROFESSIONAL AFFILIATIONS

Society of Rheology, American Institute of Chemical Engineers